RSA

Mechanics of the RSA Algorithm (Step 1 Public Key)

Public Key

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GENERATE PUBLIC KEY

Private Key

Mechanics of the RSA Algorithm (Step 1 Public Key)

Select Two Primes P & Q

$P = 53$ $Q = 59$

Start to calculate first part of the Public Key = n

$PQ = n$ Or $53 \, x \, 59 = 3127$

We also need a small exponent e

e must:

1) Be an Integer
2) Not be a factor of n
3) $1 < e < \varphi(n)$
   For this example $e = 3$

Our public key is made up of n & e or 3127 & 3

Mechanics of the RSA Algorithm (Step 2 Private Key)

We need to calculate $\varphi(n)$

$$\varphi(n) = (P-1)(Q-1)$$

$$\varphi(n) = (53-1)(59-1)$$

$$\varphi(n) = 3016$$

Now Calculate Private Key = d

$$d = \frac{2(\varphi(n))+1}{e}$$

$$2011 = \frac{2(3016)+1}{3}$$

Private Key

$d = 2011$

Mechanics of the RSA Algorithm (Encryption with Public Key)

We now have everything we need

ENCRYPT DATA WITH PUBLIC KEY

Mechanics of the RSA Algorithm (Encryption with Public Key)

We now have everything we need

1) Our public key is made up of n & e or 3127 & 3
2) Our Private Key of d or 2011

A simple example. Lets encrypt "HI"
Convert letters to numbers: H = 8 & I = 9
"89"

c = Encrypted Data

$$c = 89^e \bmod n$$

$$1394 = 89^3 \ mod \ 3127$$

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DECRYPT DATA WITH PRIVATE KEY

Mechanics of the RSA Algorithm (Decryption with Private Key)

Decrypt the data with the Private Key

c = Encrypted Data
d = Private Key
n = Public Key

Decrypted Data = $c^d \, mod \, n$
$89 = 1394^{2011} \, mod \, 3127$

Digital Signatures

- Digital signatures attempt to guarantee the identity of the person sending data from one point to another.

Digital Signatures

- The basic design of a digital signature is as follows:
  - A small amount of data is encrypted using a prescribed algorithm.
  - That small amount of data is bound to the core data being sent as well as to the identity of the sender (often a public key).

Digital Signatures

- This provides the authentic signature with the key so the recipient can verify the digital signature on any data sent with that signature, thus ensuring that the data is unmodified and sent by the authentic sender.

Digital Signatures

- When creating a digital signature, the software being used to generate the signature sends the data being transferred through a hash algorithm.

Digital Signatures

- After the data is sent through the hash, an output is generated in the form of a fixed-size chunk of data called a message digest. The message digest is unique for every chunk of data passed through the hash algorithm.

Digital Signatures

- A digital signature hash algorithm should be able to guarantee the following:
  - The original data cannot be re-created out of the message digest, and vice versa.

Digital Signatures

- There should never be replication of a message digest between two separate files because the algorithm used should create entirely different end results when the data passes through the hash.
- Message digest should always be generated by using the entire piece of data being sent.